

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with hands-on experience in Verilog, Python, and FPGA hardware design. Adept at designing and developing innovative projects that enhance system performance and reliability. Committed to continuous learning and contributing to dynamic teams focused on cutting-edge technology solutions.

EXPERIENCE

Intern

Essar Power Hazira LTD, Surat, India

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems integral to boiler-actuated power generation, contributing to a 15% increase in operational efficiency.
- Conducted in-depth research on link mechanisms for transporting waste loads from boilers, leading to a proposed design that could reduce waste handling time by 20%.
- Monitored and recorded gauge and probe data, ensuring safety compliance and enhancing operational protocols through active participation in safety seminars.

Intern

3D Cubic, Surat, India

Aug 2021 - Oct 2021

- Completed a comprehensive internship focused on 3D printing technology, where I successfully designed and fabricated a polyamide cube measuring 2 cm x 2 cm x 2 cm, demonstrating precision in additive manufacturing processes.
- Collaborated with a team to optimize 3D printing workflows, resulting in a 10% reduction in material waste during production.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, FPGA Design, PCIe, C, MATLAB, Cadence, Synopsys VCS, Primetime, Quartus II IDE, Proteus, P-spice, G-Code, Signal Generators, Oscilloscopes, DMM, Sensors, Probes, Bread Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed comprehensive block and state diagrams for PCI to SRAM transactions, achieving a design that supports 1.5 Gbps data transfer rates.
- Prepared detailed timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring compatibility and performance optimization.

Vending Machine System

Aug 2023 - Dec 2023

- Integrated multiple finite state machines, including idle, selection, payment, dispensing, and reset states, ensuring a seamless user experience.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 95% success rate in hardware validation.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Achieved a significant reduction in glitch noise by adding two bits, resulting in expected theoretical and simulated values of 12V with 99% accuracy.
- Conducted extensive testing to validate performance metrics, confirming design resilience under varying operational conditions.

STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints scripts to generate comprehensive setup and hold timing reports, identifying and rectifying timing violations that improved overall design performance by 30%.
- Optimized generated slack, significantly enhancing the reliability and efficiency of the microprocessor design.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented a verification module using Perl scripts to ensure the functionality of the adder, achieving a 100% accuracy rate in timing reports.
- Generated detailed timing reports by providing design constraints, enhancing the static timing analysis process and achieving optimal performance.